

Introduction to Lie Group Theory

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Summer 2026

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Chapter 1

Matrix Groups

1.1 Groups of Matrices

Let $M_{m,n}(\mathbb{K})$ be the set of $m \times n$ matrices whose entries are in $\mathbb{K}$, a commutative field. We will also use the special notations $M_n(\mathbb{K}) = M_{n,n}(\mathbb{K})$, and $\mathbb{K}^n = M_{n,1}(\mathbb{K})$.

$M_{m,n}(\mathbb{K})$ is a vector space over $\mathbb{K}$ with dimension mn . As well as being a vector space over $\mathbb{K}$ with dimension n^2 , $M_n(\mathbb{K})$ is also a ring with the usual addition and multiplication of matrices. It is also an important example of a finite dimensional algebra over $\mathbb{K}$.

We will use the notation

$$GL_n(\mathbb{K}) = \{A \in M_n(\mathbb{K}) : \det A \neq 0\}$$

for the set of invertible $n \times n$ matrices, known as the $n \times n$ *general linear group*, and

$$SL_n(\mathbb{K}) = \{A \in M_n(\mathbb{K}) : \det A = 1\} \subseteq GL_n(\mathbb{K})$$

for the set of $n \times n$ unimodular matrices, known as the $n \times n$ *special linear group*.

Theorem 1.1. *The sets $GL_n(\mathbb{K})$ and $SL_n(\mathbb{K})$ are groups under matrix multiplication. Furthermore, $SL_n(\mathbb{K})$ is a subgroup of $GL_n(\mathbb{K})$.*

Since determinant of a matrix is a polynomial in the entries of the matrix, it can be proved that the determinant function

$$\det : M_n(\mathbb{K}) \longrightarrow \mathbb{K}$$

is continuous.

Proposition 1.1. Let $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$. Then

- i) $GL_n(\mathbb{K}) \subseteq M_n(\mathbb{K})$ is an open subset;
- ii) $SL_n(\mathbb{K}) \subseteq M_n(\mathbb{K})$ is a closed subset.

Proof. $GL_n(\mathbb{K}) = M_n(\mathbb{K}) - \det^{-1}\{0\}$, which is open since $\{0\}$ is closed. Similarly, $SL_n(\mathbb{K}) = \det^{-1}\{1\}$, which is closed since the singleton $\{1\}$ is closed in $\mathbb{K}$. ■

Definition 1.1. Let G be a topological space, and view $G \times G$ as the product space. Suppose that G is also a group with multiplication map $\text{mult} : G \times G \rightarrow G$, and inverse map $\text{inv} : G \rightarrow G$. G is a *topological group* if mult and inv are continuous.

Theorem 1.2. For $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$, $GL_n(\mathbb{K})$ and $SL_n(\mathbb{K})$ are topological groups.

1.2 Matrix Groups

Definition 1.2. A subgroup G of $GL_n(\mathbb{K})$ is called a *matrix group* if it is a closed subset of $GL_n(\mathbb{K})$.

Remark. We may consider $GL_n(\mathbb{K})$ as a subgroup of $GL_{n+1}(\mathbb{K})$ by identifying the $n \times n$ matrix $A = [a_{ij}]$ with

$$\begin{bmatrix} A & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} a_{11} & \cdots & a_{1n} & 0 \\ \vdots & \ddots & \vdots & \vdots \\ a_{n1} & \cdots & a_{nn} & 0 \\ 0 & \cdots & 0 & 1 \end{bmatrix}$$

And it is easily verified that $GL_n(\mathbb{K})$ is closed in $GL_{n+1}(\mathbb{K})$. With the aid of this embedding, any matrix subgroup of $GL_n(\mathbb{K})$ can also be viewed as a matrix subgroup of $GL_{n+1}(\mathbb{K})$.

Definition 1.3. Let G, H be two matrix groups. A groups homomorphism $\varphi : G \rightarrow H$ is a *continuous homomorphism of matrix groups* if it is continuous and its image $\varphi(G)$ is a closed subgroup of H .

Examples. Some important examples of real and complex matrix groups:

1. Groups of Upper Triangular Matrices

For $n \geq 1$, an $n \times n$ matrix $A = [a_{ij}]$ is said to be *upper triangular* if $a_{ij} = 0$ if $i < j$. An upper triangular matrix is said to be *unipotent* if all of its diagonal entries are 1.

The *upper triangular subgroup* or *Borel subgroup* of $GL_n(\mathbb{K})$ is

$$UT_n(\mathbb{K}) = \{A \in GL_n(\mathbb{K}) : A \text{ is upper triangular}\},$$

while the *unipotent subgroup* of $GL_n(\mathbb{K})$ is

$$SUT_n(\mathbb{K}) = \{A \in GL_n(\mathbb{K}) : A \text{ is unipotent}\}.$$

It is easy to verify that $UT_n(\mathbb{K})$ and $SUT_n(\mathbb{K})$ are closed subgroups of $GL_n(\mathbb{K})$, and that $SUT_n(\mathbb{K})$ is a subgroup of $UT_n(\mathbb{K})$.

2. Affine Groups

The *n-dimensional affine group* over $\mathbb{K}$ is

$$Aff_n(\mathbb{K}) = \left\{ \begin{bmatrix} A & t \\ 0 & 1 \end{bmatrix} : A \in GL_n(\mathbb{K}), t \in \mathbb{K}^n \right\}$$

which is clearly a closed subgroup of $GL_{n+1}(\mathbb{K})$. If we identify $x \in \mathbb{K}^n$ with $\begin{bmatrix} x \\ 1 \end{bmatrix} \in \mathbb{K}^{n+1}$, then as a consequence of

$$\begin{bmatrix} A & t \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ 1 \end{bmatrix} = \begin{bmatrix} Ax + t \\ 1 \end{bmatrix}$$

we obtain an action of $Aff_n(\mathbb{K})$ on $\mathbb{K}^n$.

The group $Aff_n(\mathbb{K})$ can be expressed as the semi-direct product of $Trans_n(\mathbb{K})$ and $GL_n(\mathbb{K})$, where

$$Trans_n(\mathbb{K}) = \left\{ \begin{bmatrix} I_n & t \\ 0 & 1 \end{bmatrix} : t \in \mathbb{K}^n \right\}$$

is a closed normal subgroup of $Aff_n(\mathbb{K})$. That is,

$$Aff_n(\mathbb{K}) = GL_n(\mathbb{K}) \ltimes Trans_n(\mathbb{K}).$$

3. Orthogonal Groups

The *n × n real orthogonal group* is the subset

$$O(n) = \{A \in GL_n(\mathbb{R}) : A^T A = I_n\},$$

where A^T is the transpose of $A = [a_{ij}]$ whose entries are given by $(A^T)_{ij} = a_{ji}$. $O(n)$ is a closed subgroup of $GL_n(\mathbb{R})$. For all $A \in O(n)$, since $\det A = \det A^T$, $\det A = \pm 1$. The subgroup

$$SO(n) = \{A \in O(n) : \det A = 1\}$$

is known as the *special orthogonal group*. One of the main reasons for the study of orthogonal groups is due to their relation with isometries. If $A \in GL_n(\mathbb{R})$, then it can be proved that A is a linear isometry iff A is orthogonal. The elements of $SO(n)$ are called *direct isometries* or *rotations*.

4. Unitary Groups

The $n \times n$ *unitary group* is the subset

$$U(n) = \{A \in GL_n(\mathbb{C}) : A^*A = I_n\},$$

where A^* is the *hermitian conjugate* of A , that is $A^*)_{ij} = a_{ji}$. $U(n)$ is clearly a closed subgroup of $GL_n(\mathbb{C})$. The *special unitary group* is the subset

$$SU(n) = \{A \in U(n) : \det A = 1\},$$

which is also a normal subgroup of $U(n)$.

5. Symplectic Groups

A matrix S is called *skew symmetric* if $S^T = -S$. The standard example of this is built up using the 2×2 block

$$J = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}.$$

When $m \geq 1$ we have the non degenerative skew symmetric matrix

$$J_{2m} = \begin{bmatrix} J & O_2 & \cdots & O_2 \\ O_2 & J & \cdots & O_2 \\ \vdots & \vdots & \ddots & \vdots \\ O_2 & O_2 & \cdots & J \end{bmatrix}.$$

The matrix group

$$\text{Symp}_{2m}(\mathbb{R}) = \{A \in GL_{2m}(\mathbb{R}) : A^T J_{2m} A = J_{2m}\}$$

is called the $2m \times 2m$ *real symplectic group*. It can be proved that if $A \in \text{Symp}_{2m}(\mathbb{R})$, then $\det A = 1$.

1.3 Complex Matrices as Real Matrices

Similar to how complex numbers can be viewed as a 2-dimensional real vector space with basis $(1, i)$, every $n \times n$ complex matrix can also be viewed as a $2n \times 2n$ real matrix using the following constructions or the ones similar to it.

We identify each complex number $z = x + iy$ with a 2×2 matrix by defining a function

$$\rho : \mathbb{C} \longrightarrow M_2(\mathbb{R}); \quad \rho(x + iy) = \begin{bmatrix} x & -y \\ y & x \end{bmatrix}.$$

This can also be expressed as

$$\rho(x + iy) = xI_2 - yJ.$$

As ρ is an injective ring homomorphism, we can view $\mathbb{C}$ as a subring of $M_2(\mathbb{R})$. Notice that $\rho(\bar{z}) = \rho(z)^T$.

For a $n \times n$ complex matrix Z , we define a function $\rho_n : M_n(\mathbb{C}) \longrightarrow M_{2n}(\mathbb{R})$;

$$\rho_n(Z) = \begin{bmatrix} \rho(z_{11}) & \rho(z_{12}) & \cdots & \rho(z_{1n}) \\ \rho(z_{21}) & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \vdots \\ \rho(z_{n1}) & \cdots & \cdots & \rho(z_{nn}) \end{bmatrix}$$

Then ρ_n is an continuous injective ring homomorphism. Generally for $n \times n$ real matrices X and Y ,

$$\rho_n(X + iY) = \rho_n(X) - J_{2n}\rho_n(Y)J_{2n}.$$

1.4 Matrix Groups of Normed Vector Spaces

1.5 Matrix Exponential and Logarithm

1.6 One Parameter Subgroups in Matrix Groups

Chapter 2

Tangent Spaces and Lie Algebras

2.1 Lie Algebras

2.2 Curves and Tangent Spaces

2.3 Lie Algebras of Matrix Groups

2.4 $SO(3)$ and $SU(2)$

2.5 Complexification of Real Lie Algebra

Chapter 3

Lie Groups

- 3.1 Smooth Manifolds**
- 3.2 Tangent Spaces and Derivatives**
- 3.3 Lie Groups**
- 3.4 Matrix Groups are Lie Groups**
- 3.5 Not All Lie Groups are Matrix Groups**

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